



RESEARCH ARTICLE

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Key Points:

- The model estimates the daily probabilities of sowing and harvesting in the year 2000
- Winter and spring wheat and rainfed versus irrigated conditions are differentiated
- Our findings have implications to improve modeling multicropping patterns

Supporting Information:

- Supporting Information S1

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Modeling the Global Sowing and Harvesting Windows of Major Crops Around the Year 2000

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Abstract The lack of spatially detailed crop calendars is a significant source of uncertainty in modeling, monitoring, and forecasting crop production. In this paper, we present a rule-based model to estimate the sowing and harvesting windows of major crops over the global land area. The model considers field workability due to snow cover and heavy rainfall in addition to crop biological requirements for heat, chilling, and moisture. Using daily weather data for the period 1996–2005 as model input, we derive calendars for maize, rice, winter and spring wheat, and soybeans around the year 2000 with a spatial resolution of 0.5° in latitude and longitude. Separate calendars for rainfed and irrigated conditions and three representative varieties (short-, medium- and long-season varieties) are estimated. The daily probabilities of sowing and harvesting derived using the model well capture the major characteristics of reported calendars. Our modeling reveals that field workability is an important determinant of sowing and harvesting dates and that multicropping patterns influence the calendars of individual crops. The case studies show that the model is capable of capturing multicropping patterns such as triple rice cropping in Bangladesh, double rice cropping in the Philippines, winter wheat-maize rotations in France, and maize-winter wheat-soybean rotations in Brazil. The model outputs are particularly valuable for agricultural and hydrological applications in regions where existing crop calendars are sparse or unreliable.

Plain Language Summary This manuscript describes a numerical model to estimate location-specific sowing and harvesting dates of crops over the globe. Ten-year-long daily weather data and a few coefficients that represent the physiological characteristics of a crop (for instance, the amount of water needs to complete the life cycle of an annual crop) are only inputs to the model. Comparisons with the reported crop calendars indicate that the model well reproduces calendars of maize, rice, winter and spring wheat, and soybean around the year 2000 in major crop-producing countries. We also find that snow cover and heavy rainfall, which influence field workability but have not considered in earlier modeling, are important to estimate sowing and harvesting dates and multicropping patterns (for instance, a combination of winter and summer crops) affect the calendar of individual crops. Our findings are useful when simulating the responses of crop calendars to climate change.

1. Introduction

Two global data sets of crop calendars have been widely used within crop models during the past years: the SAGE data set (Sacks et al. (2010) and MIRCA2000 (Portmann et al., 2010). Both data sets are based on national and subnational crop progress reports and provide crop-specific planting and harvesting dates circa the year 2000 at a 5-min resolution in latitude and longitude. Another data set derived from satellite remote sensing (Kotsuki & Tanaka, 2015) has not been widely used thus far in crop modeling because it only provides calendars for a single dominant crop per location, even if multiple crops are grown. A relatively detailed data set has only recently been developed for rice (Laborte et al., 2017). Therefore, SAGE and MIRCA2000 constitute the main sources of crop calendars available today for analyzing major crops at the global scale.

In the SAGE data set, average planting and harvesting windows for 19 crops, including major crops (maize, rice, wheat, and soybean), around the 1990s and early 2000s are reported. Different growing seasons of a crop, such as spring and winter wheat and main- and second-season maize and rice, are available in SAGE, but this distinction is not available in MIRCA2000. Instead, average planting and harvesting months for 23 crops, including the major crops, in 1998–2002 are available for each of the irrigated and rainfed conditions in MIRCA2000. However, MIRCA2000 only has monthly resolution whereas SAGE has daily

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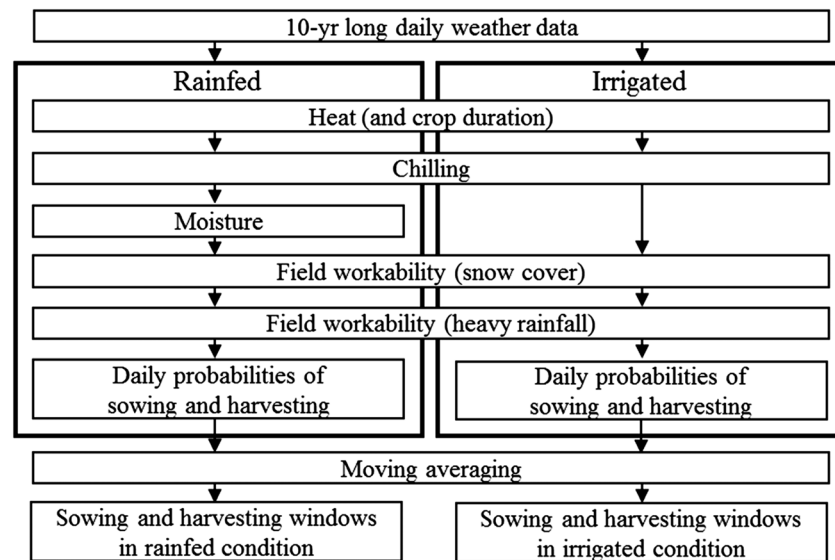


Figure 1. Schematic illustration of the sowing and harvesting windows model presented in this study.

resolution. The spatial coverage of SAGE is relatively limited: 76% for wheat, 82% for rice, 88% for maize, and 92% for soybeans (Sacks et al., 2010). MIRCA2000 mostly covers the global cropland if rainfed and irrigated conditions are combined, but data in some regions are derived by spatial extrapolation (Portmann et al., 2010), which sometimes leads to discrepancies with other data sources. Discrepancies have been reported for both data sets and include the planting month of wheat in Russia in MIRCA2000, which is extrapolated from the data in Ukraine, Norway, and Romania (Waha et al., 2012), and the misclassification of spring wheat in India as winter wheat in SAGE (Mathison et al., 2017).

Although the two data sets have the relative advantages and limitations described above, the lack of spatially explicit crop calendars is their common pitfall. In SAGE and MIRCA2000, a single value of planting date (or harvesting date) is assigned to all 5-min grid cells within an administrative unit (a “paint-by-number” approach). However, crop progress reports used to compile the data sets more strongly represent the data in major producing areas than in minor producing areas within that unit. This shortfall poses a challenge to test and improve the management components (sowing date model in particular) within crop models (Elliott et al., 2015; Mathison et al., 2017; Waha et al., 2012). Given the common limitation of the two major data sets, in this study, we propose a rule-based model that determines crop- and location-specific sowing and harvesting windows globally with a spatial resolution of 0.5°. The model outputs were compared with SAGE and MIRCA2000 data sets and other data sources. Then, we studied global patterns of the modeled planting and harvesting windows in irrigated and rainfed conditions for maize, rice, winter wheat, spring wheat, and soybeans around the year 2000. For each crop, calendars for three hypothetical varieties—short-, medium-, and long-season varieties—were considered.

2. Materials and Methods

2.1. Sowing and Harvesting Windows Model

The model determines the potential sowing and harvesting dates of a crop of interest at a given location based on the sequence of rules and a 10-year long daily weather data (Figure 1). The rules describe climatic requirements in terms of heat, chilling, moisture, and field workability. Rainfed and irrigated conditions are separately considered. The potential sowing date indicates days with a higher probability of completing the annual crop life cycle under a given weather pattern and soil hydrologic condition. The model outputs are moving-averaged to address sowing and harvesting windows. A sowing (harvesting) window indicates the time interval with nonzero probability of sowing (harvesting). The model formulation was identical across the Northern and Southern Hemispheres and across the low and middle-to-high latitudes; differences in model output are driven purely by variations in climatic conditions and soil profiles.

Table 1
Crop- and Variety-Specific Parameter Values Used in the Model

Crop	Variety	Parameters						
		a (—)	b (—)	T_b (°C)	T_u (°C)	Duration (days)	Chilling (°C days)	W_{needs} (mm)
Maize	Short	2,392	4,718	8	30	80–220	$CDD_c = 0$	500
	Medium	2,540	3,991					650
	Long	2,688	3,264					800
Rice	Short	2,739	7,189	8	37	80–220	$CDD_c = 0$	450
	Medium	3,135	5,458					575
	Long	3,531	3,728					700
Winter wheat	Short	3,805	5,484	0	26	120–340	$CDD_c > 0$ and $CDD_a < 3,248$	450
	Medium	4,203	4,629					550
	Long	4,601	3,774					650
Spring wheat	Short	3,805	5,484	0	26	110–160	$CDD_c \geq 0$ and $CDD_a < 3,248$	450
	Medium	4,203	4,629					550
	Long	4,601	3,774					650
Soybean	Short	2,367	10,886	10	30	90–220	$CDD_c = 0$	450
	Medium	3,204	7,459					575
	Long	4,041	4,031					700

2.1.1. Rules to Determine Potential Sowing and Harvesting Dates

The model is composed of the following sequence of rules:

1. Evaluate if the heat requirement is satisfied. For a given location, the model calculates the 10-year average annual growing-degree days (GDD) with a base temperature of 0 °C (GDD_a ; Figure S1) and computes the crop total heat requirement (GDD_c) using the relationship between GDD_a and GDD_c found in Deryng et al. (2011) (Figure S2):

$$GDD_c = a \left[1 - \exp\left(\frac{-GDD_a}{b}\right) \right], \quad (1)$$

where GDD_c is the crop total heat requirement (°C days), GDD_a is the 10-year average annual GDD (°C days), and a and b are empirical coefficients (Table 1). GDD_c increases and approaches saturation with an increase in GDD_a (Figure S2); a similar relationship is reported in van Bussel et al. (2015).

The model calculates GDD from a given candidate of potential sowing dates according to Deryng et al. (2011):

$$GDD_i = \sum_{sowing}^i \max\{0, [\min(T_i, T_u) - T_b]\}, \quad (2)$$

where the suffix i indicates the i th day after sowing, GDD is the growing-degree days (°C days), T is the daily mean temperature (°C), and T_u and T_b are the maximum and base temperatures for crop growth (°C), respectively (Table 1). The model evaluates if the heat requirement is satisfied ($GDD_i \geq GDD_c$) within a certain number of days from the given date (e.g., 220 days for maize; see Table 1). The model also checks if the crop duration (sowing to harvesting) exceeds a certain number of days to avoid unrealistically short crop durations (e.g., 80 days for maize; Table 1). These threshold criteria for duration were determined based on the data reported in U.S. Department of Agriculture (USDA, 1994). If the heat and/or duration requirement is not satisfied, the given date is deemed to be unsuitable for sowing due to the lack of warmth or unreasonable crop duration.

2. Evaluate if the chilling requirement is satisfied. For maize, rice, and soybean, $CDD_c = 0$ is required to ensure no frost day during the crop duration:

$$CDD_c = \sum_{sowing}^{harvesting} |\min(0, [T_i - 5])|, \quad (3)$$

where CDD_c is the chilling-degree days (CDD) from sowing to harvesting (°C days). For winter wheat, $CDD_c > 0$ and $CDD_a < 3,248$ are required, where CDD_a is the 10-year average annual CDD (°C days);

Figure S1). For spring wheat, $CDD_c \geq 0$ and $CDD_a < 3,248$ are used instead. A certain level of chilling (a non-zero CDD_c value) is essential for winter wheat because of vernalization, whereas it is not necessarily required for spring wheat ($CDD_c \geq 0$; Table 1). The criterion for CDD_a was determined as the 95-percentile value of the grid-cell CDD_a data over the global wheat harvested area (Figure S3). A base temperature of 5 °C was commonly used for the calculation of CDD_a and CDD_c across the crops. The base temperature value used here falls within the range found in an earlier study examining vernalization of wheat (−4 to 17 °C; van Bussel et al., 2015). If the chilling requirements are not satisfied, the given date is deemed to be unsuitable for sowing due to coldness or, for winter wheat, the lack of lower temperatures for vernalization.

3. Evaluate if the moisture requirement is satisfied. The total water supply to a crop from sowing to harvesting is calculated as follows:

$$W = \theta_0 + \sum_{\text{sowing}}^{\text{harvesting}} \{ \max(0, [\theta_i - \theta_{i-1}]) + E_i \}, \quad (4)$$

where W is the total water supply (mm); θ_0 , θ_i , and θ_{i-1} are the plant-extractable root-zone soil moisture just before sowing, for the i th day after sowing, and for the previous day, respectively (mm); and E is the actual evapotranspiration of the reference crop (mm/day). If we consider a simple water balance equation ($\theta_i - \theta_{i-1} = P_i - R_i - E_i$), where P is the precipitation (mm/day) and R is the runoff (mm/day) for explanatory purposes, precipitation minus runoff ($\theta_i - \theta_{i-1} + E_i = P_i - R_i$) represents the available water for crops for the i th day in rainfed conditions. Soil moisture just before sowing (θ_0) is also a part of the available water for crops. The amount of water required to complete the crop life cycle (W_{needs} ; mm) was obtained from Brouwer and Heibloem (1986) (Table 1). Although W_{needs} values used here are based on field observations, values may differ by location and other factors. However, no information is available to account for this uncertainty. If the total water supply is less than the water needs ($W < W_{\text{needs}}$), the moisture requirement is not satisfied, and the given date is deemed to be unsuitable for sowing due to the lack of water in rainfed conditions. In irrigated conditions, the model uses the assumption that this requirement is always satisfied (Figure 1).

4. Evaluate field workability and growing conditions based on snow cover. This requirement was included in the model with reference to Trnka et al. (2011). For all crops except winter wheat, there can be no snow cover during the entire period of crop growth, as well as 1 day prior, to account for field workability. For winter wheat, snow cover can be present during the growing period, except during the days of sowing and harvesting and 1 day prior to sowing/harvest days, to account for field workability. If the requirements are not satisfied, the given date is deemed to be unsuitable for sowing due to the poor field workability associated with snow cover.
5. Evaluate if there is no heavy rainfall (>10 mm/day) on the day of sowing and 1 day prior to it to consider the field workability at sowing. No heavy rainfall at sowing is also important to avoid washout of sown seeds. The model also checks if there is no heavy rainfall on the day of harvest and 1 day before, as in Trnka et al. (2011). The absence of heavy rainfall around harvesting is important not only for the field workability but also to avoid preharvest sprouting. If the requirements are not satisfied, the given date is deemed to be unsuitable for sowing due to the poor field workability associated with heavy rainfall.
6. If all requirements mentioned above are satisfied, the given date is deemed as a potential sowing date for that location, and the simulated date of harvesting is deemed as a potential harvesting date.

We conducted a 10-year (1996–2005) model simulation for each day of the year from 1 January to 31 December and derived the daily probability of completing the annual crop life cycle around the year 2000 (i.e., the probability of sowing; p_{sow}) and the corresponding daily probability of harvesting (p_{harvest}). A 5-day running average was applied 10 times to the daily probabilities of sowing (harvesting) with the periodic boundary condition to help visually address sowing (harvesting) windows.

2.1.2. Daily Weather Data

The daily mean temperature and precipitation data were obtained from the 56-year (1958–2013), 0.5°-resolution, global retrospective meteorological forcing data set (S14FD; Iizumi, Takikawa, et al., 2017). The daily actual evapotranspiration of the reference crop, root-zone soil moisture, and snow water-equivalent data was derived by running a soil water balance model coupled with the Penman-Monteith equation and a snow cover model using S14FD forcing and plant-extractable water holding capacity of the soil as the inputs. These soil water balance and snow cover models are described in Iizumi, Furuya, et al. (2017). A 37-year (1958–1994) spin-up simulation was conducted before the data for the studied period were derived.

Table 2
The Contingency Table Used When Calculating the Skill Score

		Observed	
		Within window	Outside window
Prediction	Within window	A	B
	Outside window	C	D

2.2. Model Evaluation

2.2.1. Reported Crop Calendar Data

We compared the SAGE and MIRCA2000 data sets with the modeled sowing and harvesting windows. As already noted, a single value was assigned to all grid cells within an administrative unit in SAGE and MIRCA2000. We assumed that calendars in these data sets represent those in major producing areas in that unit (e.g., country). Therefore, for each major producing country, we selected a single grid cell that

had the largest rainfed area of a crop of interest using MIRCA2000. Similarly, locations with the largest irrigated area were selected (with the caveat that effective data for the selected locations were sometimes lacking in these data sets; Figures S4, S5, S6, and S7). The reported planting and harvesting windows could be different across rainfed and irrigated conditions when the selected grid cells for rainfed and irrigated conditions within a country were different.

Additionally, we compared our model simulations to other reported calendars for multicropping regions: triple rice cropping in Bangladesh (Gray et al., 2014; USDA, 1994); double rice cropping in the Philippines (Koide et al., 2013); crop rotation of winter wheat and maize in France (Loubet et al., 2011); and crop rotation of maize, winter wheat, and soybeans in Brazil (Anderson et al., 2017; Cohn et al., 2016). The global data reported in USDA (1994) were also used because clear distinctions between multiple cropping seasons were available.

2.2.2. Skill Score

The performance of the model in predicting sowing and harvesting windows was measured using the Heidke skill score (Wilks, 2006), which measures fractional improvements over random chance and is generally used for categorical predictions such as in Table 2.

Based on the contingency table mentioned above, the Heidke skill score (*HSS*) was computed as follows:

$$HSS = \frac{2(AD - BC)}{(A + C)(C + D) + (A + B)(B + D)}. \quad (5)$$

HSS = 1 indicates perfect prediction, *HSS* = 0 indicates random prediction (no skill), and *HSS* < 0 indicates that the model performance is worse than random prediction.

In this study, we assumed that a day with $p_{\text{sow}} > 0.05$ or $p_{\text{harvest}} > 0.05$ fell within the predicted sowing or harvesting window, respectively. The score values were calculated for sowing and harvesting windows relative to a single data set (SAGE or MIRCA2000) and averaged. When the SAGE data set was compared, the score values were calculated for both main and second seasons, if any, and averaged. Then, score values were averaged over the different data sets to derive an overall score value.

3. Results

3.1. Comparisons With the Reported Data

The daily probabilities of sowing and harvesting for maize and soybeans in the United States are shown in Figure 2 as examples. The modeled maize harvesting window for long-season varieties showed a better match with the reported data (USDA, 1994) than for short- and medium-season varieties, although the modeled sowing windows for all of the varieties were comparable to the reported data (Figures 2a–2c). In the United States, soybeans are often planted after maize planting is completed (Sacks & Kucharik, 2011). The model suggested that short- and medium-season varieties were more likely than long-season varieties to be consistent with the reported soybean calendars (Figures 2d–2f; in fact, the model simulated zero probability for long-season soybeans).

The results for major producing countries are shown in Figure 3. For each crop and country, the single variety that gave the highest skill score value was selected. The model well captured the timing of the reported sowing and harvesting windows for maize and soybean in the United States and China (*HSS* = 0.36–0.72; Figures 3a and 3c) and for spring wheat in these countries and Canada (0.30–0.73; an exception of 0.10 for irrigated conditions in China was due to the MIRCA2000 data set incorrectly assigning the data for winter wheat to spring wheat-grown areas; Figures 3b and 3e). However, large discrepancies were found for rice

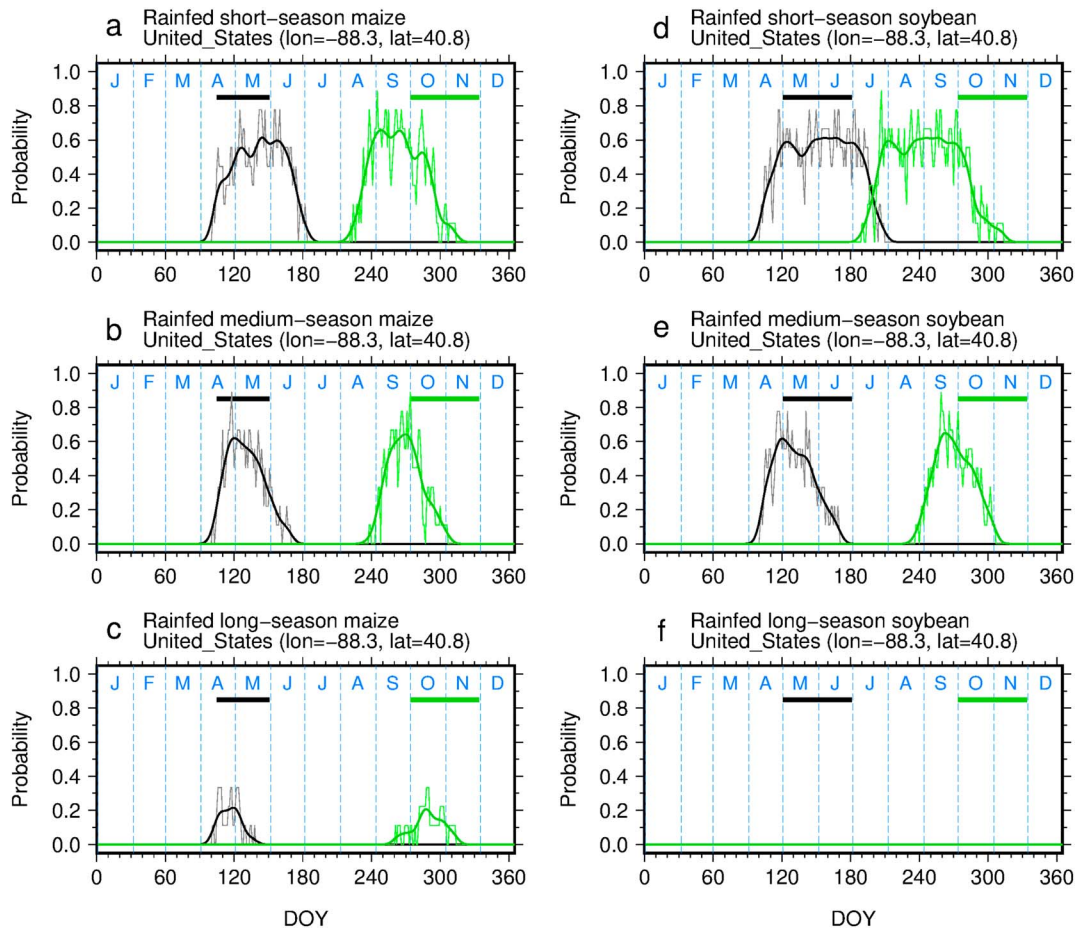


Figure 2. Comparisons between the daily probabilities of (black) sowing and (green) harvesting and reported data (USDA, 1994) for maize and soybeans in the United States. A thin line indicates the original model outputs; a solid line indicates 5-day moving average modeled data (it conducted 10 times); a horizontal line in the upper part of a panel indicates the reported (black) planting and (green) harvesting windows; and a vertical dashed line with character indicates month.

in China, India, and Bangladesh (0.00–0.19); winter wheat in China, the United States, and Canada (0.01–0.13); and maize and soybean in Brazil (0.00–0.32). For winter wheat, the overestimated length of sowing and harvesting windows explained the discrepancies (Figure 3b). However, for rice, the presence of multiple cropping and the uncertainty of the reported calendars associated with the different data sets contributed to the discrepancies.

In comparison with the reported data from USDA (1994), the modeled sowing and harvesting windows for rice in India reasonably captured both the Kharif season (the crop duration spans from May to February [Figure 4a]; rainy and thus rainfed) and the Rabi season (the crop duration spans from November to May [Figure 4b]; dry and thus irrigated). In SAGE, Kharif and Rabi seasons were classified into main and second seasons, respectively. Although the Rabi season rice is mainly operated in irrigated conditions, the presence or absence of irrigation was not identifiable in the SAGE data set. The inclusion of the data for the second (irrigated Rabi) season when evaluating the modeled calendars in rainfed conditions may explain the low skill score value for rainfed rice in India. Similarly, the inclusion of the data for the main (Kharif) season was the reason for the discrepancy for irrigated rice. The reasons for the discrepancies between modeled and reported dates for Rice in Bangladesh are discussed in the next section.

3.2. Case Studies for Multicropping Patterns

3.2.1. Triple Rice Cropping in Bangladesh

Triple rice cropping is commonly practiced in Bangladesh (Gray et al., 2014; USDA, 1994). The daily probabilities of sowing for rainfed rice showed two peaks in May and June when a medium-season variety was

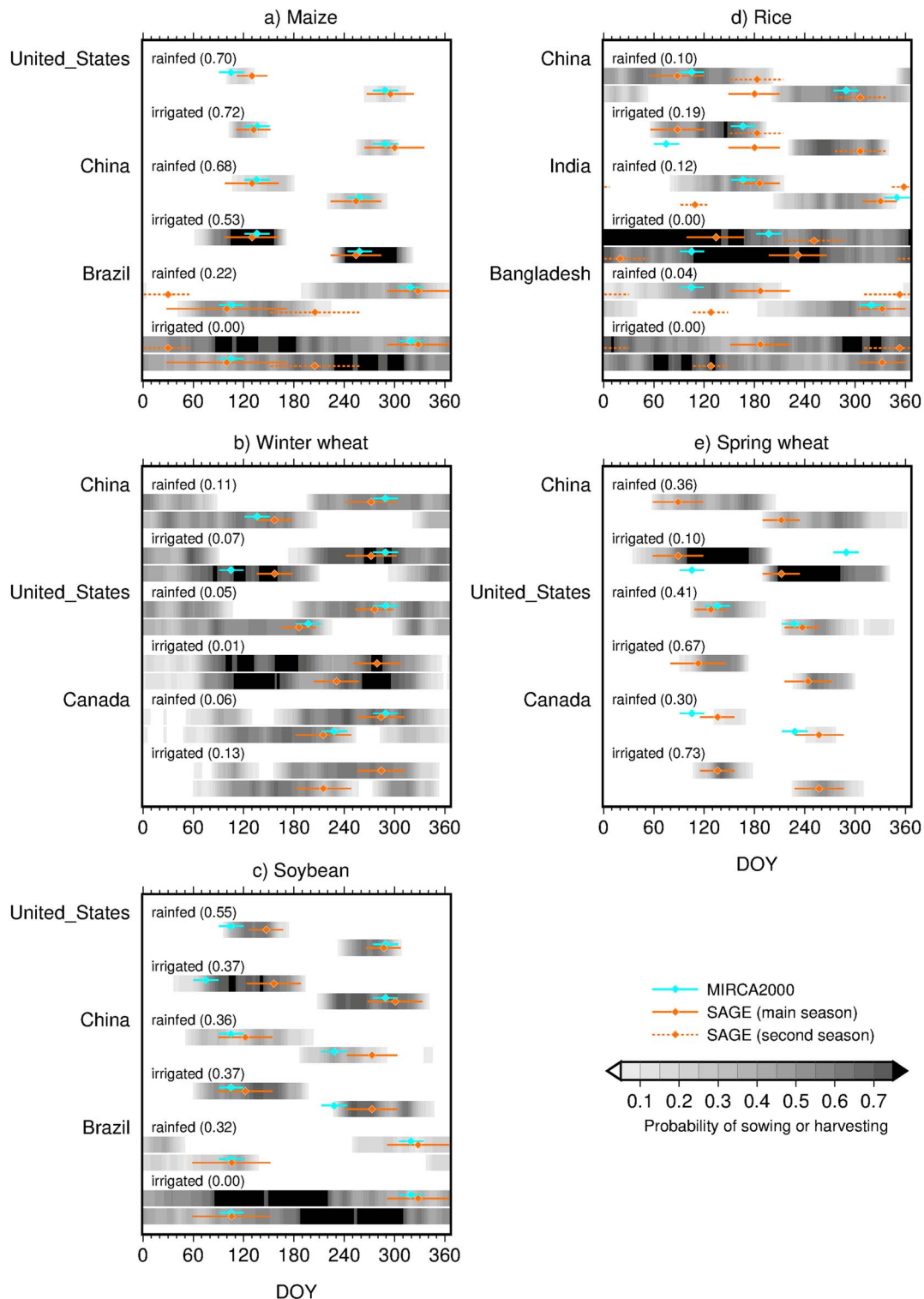


Figure 3. Comparisons between the daily probabilities of (upper) sowing and (lower) harvesting and SAGE and MIRCA2000 data sets for major crops in selected major producing countries. A number in parentheses indicates the overall skill score. $p_{\text{sow}} > 0.05$ and $p_{\text{harvest}} > 0.05$ are used as the thresholds.

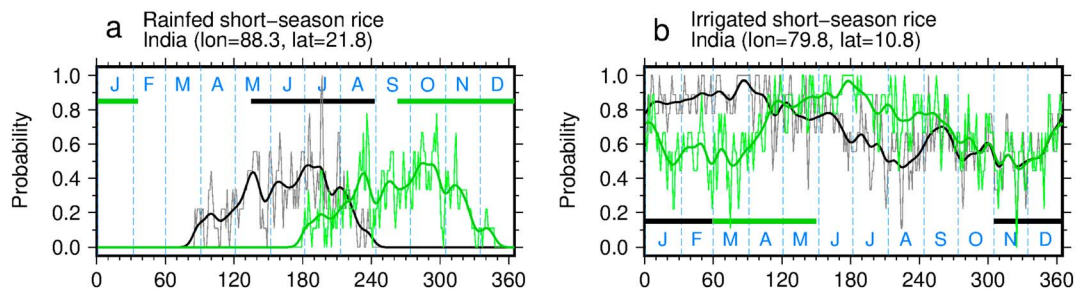


Figure 4. Same as Figure 2, but for (a) Kharif rice and (b) Rabi rice in India.

assumed (Figure 5a). The comparison with the reported planting windows (USDA, 1994) suggested that these peaks captured the Aus and Aman seasons. On the other hand, the higher probabilities of sowing for irrigated rice appeared from November to January (Figure 5b) and matched well with the reported planting window of the Boro season. The modeled and reported harvesting windows for the Boro season were comparable (Figure 5b). For the Aus and Aman seasons, the reported harvesting windows overlapped with the time intervals with nonzero probability of harvesting but not the peaks of the probability of harvesting. These results suggested that the Aus and Aman seasons are produced in rainfed conditions and the Boro season is operated under irrigated conditions.

When SAGE was compared with the modeled calendars for rice in Bangladesh, the reported data for the Aus and Aman seasons were included in the comparison for irrigated condition. The data for the Boro season were included when assessing the modeled data in rainfed conditions. These in part explained the discrepancies mentioned earlier (Figure 3d).

3.2.2. Double Rice Cropping in the Philippines

Double rice cropping is commonly practiced in the Philippines (Koide et al., 2013; USDA, 1994). The higher probabilities of sowing (harvesting) for rainfed rice emerged from near April to June (September to December), when a long-season variety was assumed (Figure 5c). On the other hand, the higher probabilities of sowing (harvesting) for irrigated rice appeared from near October to December (March to April) when a short-season variety was assumed (Figure 5d). These time intervals reasonably matched with the main and second seasons (USDA, 1994) or “rainy” July–December and “dry” January–June cropping seasons (Koide et al., 2013). These results suggest that the main and the second season are operated in rainfed and irrigated conditions, respectively, and a long-season variety is likely used in the main (rainy) season, whereas a short-season variety is likely adopted in the second (dry) season.

3.2.3. Winter Wheat and Maize Rotation in France

The model showed that if a short-season variety was assumed, summer maize was sown from approximately April to May and harvested from approximately September to November. These time intervals matched the reported calendars (USDA, 1994; Figure 5e). For winter wheat, the time intervals with higher probability of sowing and harvesting were much longer than the reported windows for planting (October to November) and harvesting (July to August; USDA, 1994; Figure 5f). However, if crop rotation was considered, that is, maize (May to September), winter wheat (October to July), Barley (October to June), and mustard (September to April) (Loubet et al., 2011), no overlap occurred between maize harvesting and winter wheat sowing. This suggests that the actual sowing windows are influenced by the harvesting windows of the preceding crop because of field management.

3.2.4. Maize, Winter Wheat, and Soybean Rotation in Brazil

In Brazil, known multicropping patterns include the traditional rotation of first-crop maize (October to May), winter wheat (May to November), soybeans (December to May), and cover crops (May to October; Anderson et al., 2017) and the recent rotation of short-season soybeans (September to February) and short-season (safrinha) maize (March to July; Anderson et al., 2017; Cohn et al., 2016). Safrinha maize is also known as second-crop winter maize (USDA, 2007). The model reasonably captured the reported calendars of the traditional pattern (Figures 5g–5j). However, for Safrinha maize, only January (June) overlapped between the modeled and reported sowing (harvesting) windows when the data of USDA (1994) were compared (Figure 5h). Discrepancies became larger when the data of Anderson et al. (2017) (crop duration from

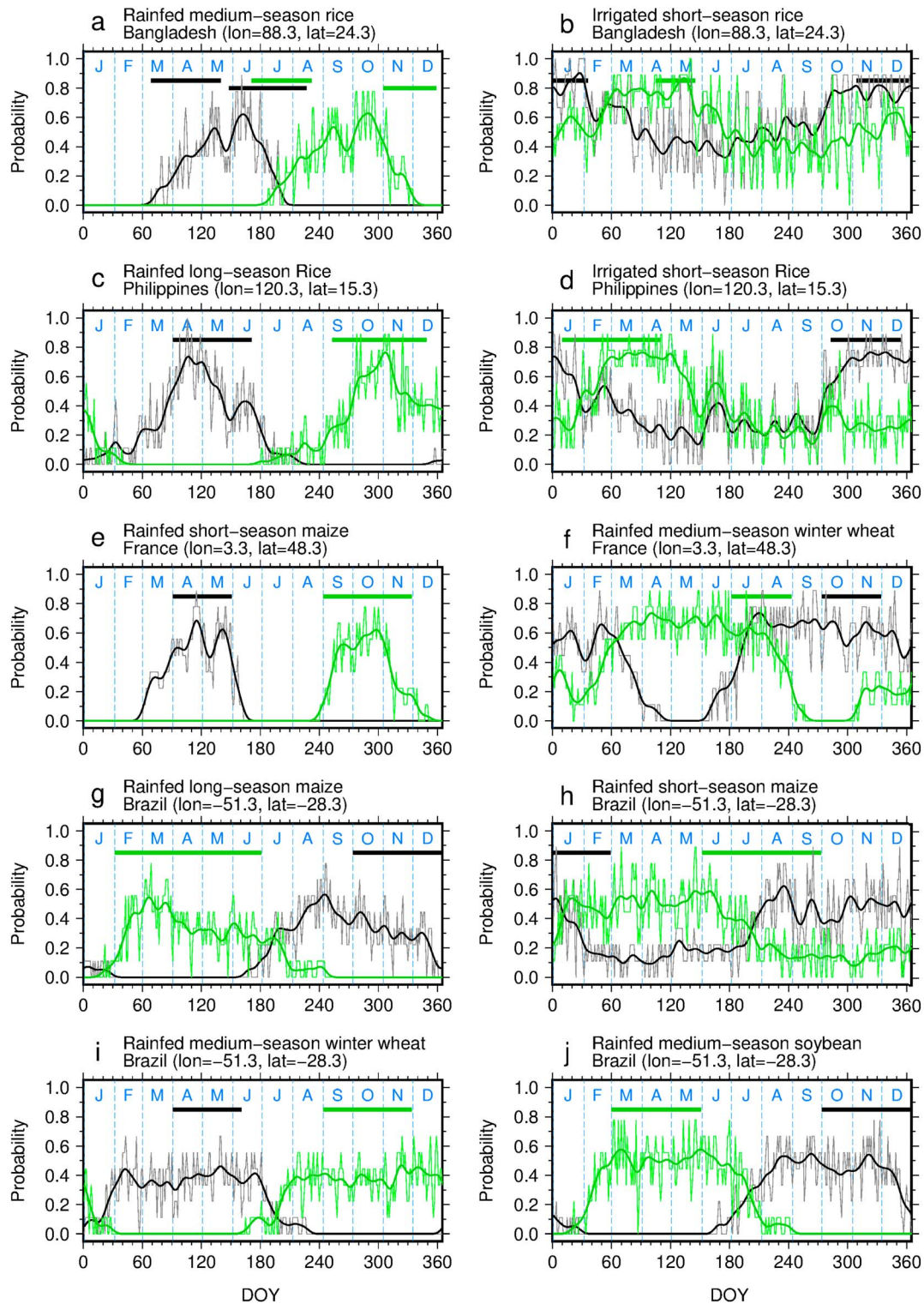


Figure 5. Same as Figure 2, but for selected worldwide multicropping regions.

March to July) were compared. Because the recent cropping pattern has rapidly expanded since the 2000s (Anderson et al., 2017; USDA, 2007) and USDA (1994) is based on older data in the 1990s, second-crop maize (USDA, 1994) may not exactly correspond to Safrinha maize. More information is necessary to explain the discrepancy more conclusively.

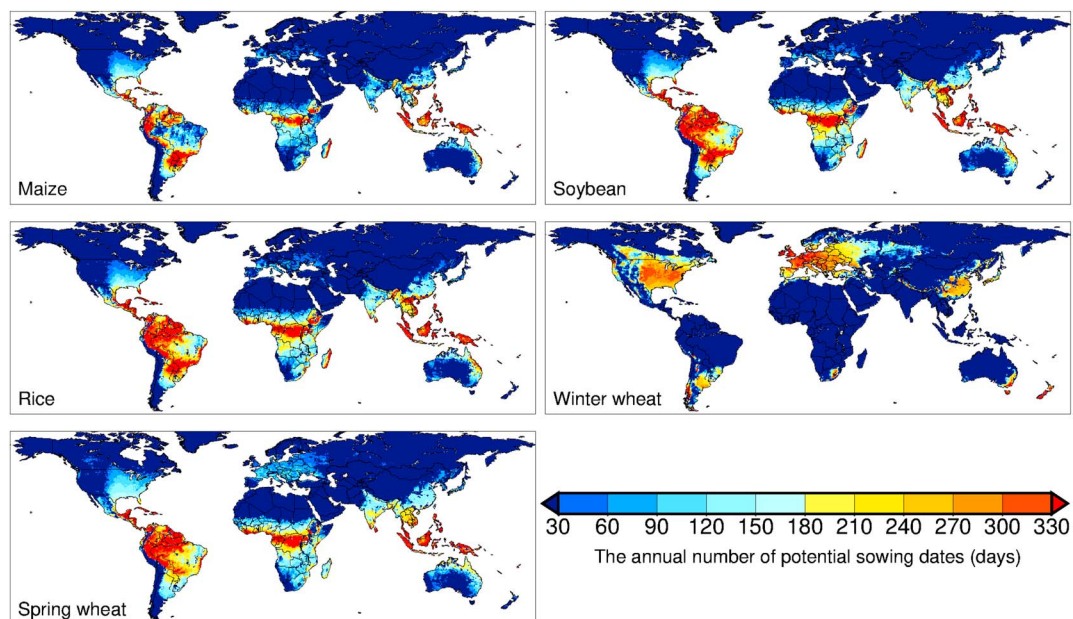


Figure 6. The annual number of potential sowing dates for medium-season variety crops in rainfed conditions. Days with $p_{\text{sow}} > 0$ are summed.

3.3. Global Patterns of Potential Sowing Dates

The model revealed clear geographical patterns in the annual number of potential sowing dates in rainfed conditions (Figure 6). The geographical patterns were largely different between winter (winter wheat) and summer crops (maize, rice, spring wheat, and soybeans). For winter wheat, many more potential sowing dates than other regions appeared around the American Midwest, the Canadian prairie, West Europe, and South China. Moreover, the Argentine Pampa, Indo-Gangetic Plains, New Zealand, and the southeastern part of Australia also showed many potential sowing dates for winter wheat (Figure 6).

Compared with the differences between the winter and summer crops, the differences across the summer crops were small. Many more potential sowing dates of the summer crops emerged in the low latitudes (equatorward of $\pm 30^\circ$). However, in the northern part of South America and the Guinean Coast in West Africa, the areas with many potential sowing dates of rice, spring wheat, and soybeans were more widespread than those of maize. This is mainly because maize requires more water to complete its life cycle than the remaining summer crops (Table 1). For all crops considered here, the potential sowing dates in irrigated conditions in many parts of the world, except for the high latitudes, were larger in number than those in rainfed conditions (Figure S8).

3.4. Dominant Climatic Constraints of Sowing

The climatic factors that dominantly govern the potential sowing dates are shown in Figure 7. Note that the dominant climatic constraints are the same between sowing and harvesting because the model used here considers crop development from sowing to harvesting (or termination of crop development due to climatic constraint). Maps of dominant climatic constraints for harvesting are thus omitted. The geographical patterns of the winter and summer crops are substantially different but are similar across the summer crops. For winter wheat, the lack of moisture appeared as the main climatic constraint in rainfed conditions in the Great Plains, Canadian prairie, Northeast China, and Russia, whereas the poor workability due to snow cover dominated in East Europe and the northern part of the American Midwest (Figure 7). The lack of lower temperature for vernalization was the main constraint for winter wheat in Mediterranean Europe, South China, the Argentine Pampa, and Australia. Two categories, no climatic constraint of sowing and poor field workability due to heavy rainfalls, were rarely found for winter wheat.

For the summer crops, the lack of warmth dominated in the middle-to-high latitudes (poleward of 30° ; Figure 7). In the low latitudes, lack of moisture was the widespread climatic constraint in rainfed

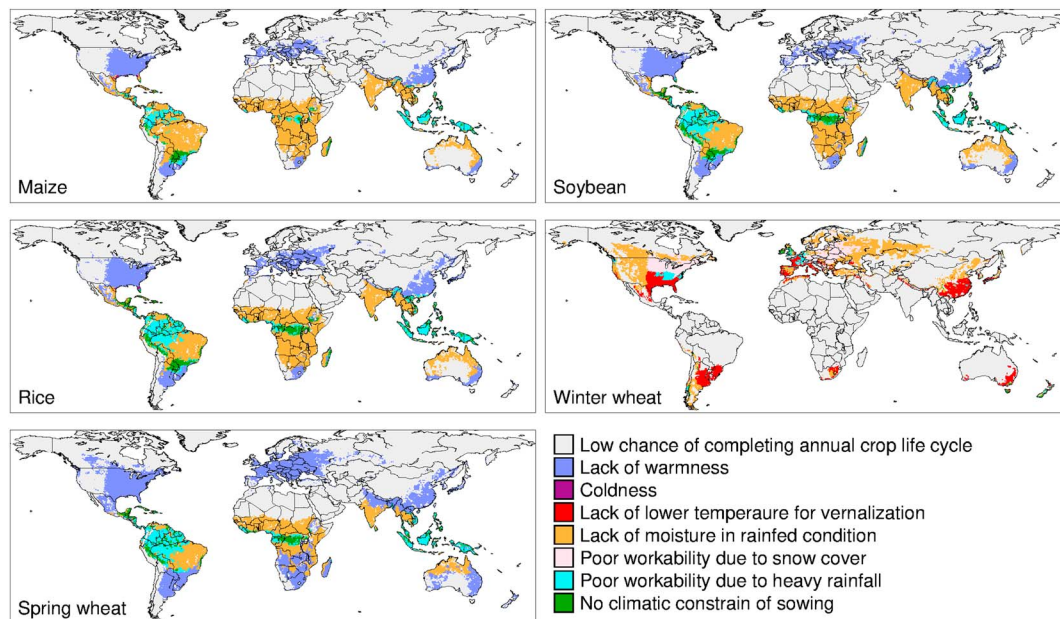


Figure 7. The main climatic constraints of sowing for the individual crops of medium-season variety in rainfed conditions. Regions where the annual number of days with $p_{\text{sow}} > 0$ is less than 10 days are labeled as “Low chance of completing annual crop life cycle,” and regions where the annual number of days with $p_{\text{sow}} > 0$ is larger than 355 days are labeled as “No climatic constrain of sowing.” In regions where the annual number of days with $p_{\text{sow}} > 0$ ranges from 10 to 355 days, a single climatic factor most frequently appeared as the constraint for sowing during the presented studied period. A climatic constraint labeled as “Lack of lower temperature for vernalization” is used only for winter wheat.

conditions. In the humid tropics, no climatic constraint of sowing or poor workability due to heavy rainfalls appeared for the summer crops.

For both the winter and summer crops, unrealistically short crop durations were rarely found thanks to the use of T_u (maximum base temperature) in the model, which limits the influences of high temperature on growth rate. Coldness rarely appeared as a constraint for both winter and summer crops because the influences of low temperature were predominantly classified as a low chance of completing the annual crop life cycle.

In irrigated conditions, the lack of lower temperatures for vernalization and the poor field workability due to snow cover dominated for winter wheat (Figure S9). For the summer crops, a lack of warmness appeared over the middle-to-high latitudes, whereas no climatic constraint or the poor workability due to heavy rainfalls dominated in the low latitudes.

4. Discussion

4.1. Advantages of the Modeled Crop Calendars

The crop calendars derived using the model have some notable characteristics. First, these cover the global land area with more spatial detail than the SAGE and MIRCA2000 data sets. The results show that the modeled calendars are comparable to the reported planting and harvesting windows in many cases. Because subnational data are only used for a few major crop-producing countries in the SAGE and MIRCA2000 data sets (Portmann et al., 2010; Sacks et al., 2010; Waha et al., 2012), the modeled calendars enable users to take spatial variations in sowing and harvesting dates into account for their specific applications. Moreover, the size of administrative units varies considerably, which indicates that the spatial representativeness of calendars in SAGE and MIRCA2000 varies by location. In contrast, the modeled calendars have more consistent representativeness across locations.

Second, distinctions are made between irrigated and rainfed conditions, winter and spring wheat, and different varieties in the modeled calendars. Calendars for double cropping are available for some crops in the SAGE data set but are not linked with the presence or absence of irrigation. Calendars for irrigated

and rainfed conditions are available in the MIRCA2000 data set, but different growing seasons of a crop are not identified. Additionally, the planting and harvesting windows of 1 month assumed in the MIRCA2000 data set are shorter than the actual windows. The classification of cropping season sometimes differs by data set. An example is the Aus and Aman rice seasons in Bangladesh between the SAGE data set and USDA (1994).

Third, the modeled calendars offer a powerful means to address the complexity of crop calendars in multicropping regions. As presented in Waha et al. (2013), the replacement of single cropping with double cropping systems could be an adaptation measure for climate change in sub-Saharan Africa. The modeled calendars provide a basis for assessing the feasibility of multicropping under warming at the global scale.

Fourth, the modeled calendars are based on a theoretical understanding of the crop's biological requirements for heat, chilling, and moisture and empirical knowledge of field workability. Therefore, these are largely independent of other data sets derived from crop progress reports (Laborte et al., 2017; Portmann et al., 2010; Sacks et al., 2010) or satellite remote sensing (Kotsuki & Tanaka, 2015) and are modeled based only on climatic conditions (Deryng et al., 2011; Mathison et al., 2017; Waha et al., 2012).

Lastly, the modeled calendars offer crop modelers opportunities to improve the realism of their crop growth simulation over a large spatial domain. If we consider a 0.5° grid cell as an example, sowing window is more realistic than single deterministic sowing date as indicated by the fact that most of the existing data sets (SAGE and USDA, 1994) explicitly report planting window. Perhaps, the daily probability of sowing derived from our model is not readily applicable to many crop models that currently require deterministic sowing date as the input. However, note that some studies already incorporate sowing (or transplanting) window that is essentially the same with the daily probability of sowing into crop model simulation (Hasegawa et al., 2008; Sawano et al., 2008).

4.2. Improvements to Sowing Date Modeling

Because the lack of high-resolution crop calendars is a significant source of uncertainty in modeling (Elliott et al., 2015), monitoring (Laborte et al., 2017; Whitcraft et al., 2015), and forecasting crop production (Iizumi et al., 2013, 2018), efforts for better understanding and modeling of sowing dates have been conducted (Choi et al., 2017; Mathison et al., 2017; Waha et al., 2012). Plausible estimates of sowing dates are key to reducing the uncertainty because dates of harvesting and other phenological events (e.g., flowering) can be accurately estimated if observed sowing dates are fed into calibrated crop phenology models (Iizumi et al., 2014; van Bussel et al., 2011).

Earlier studies have confirmed that climate is important, although not a sole determinant of sowing dates. In particular, temperature and precipitation seasonality and temperature in the coldest month play key roles in determining sowing dates in the middle-to-high latitudes and semiarid regions (Leemans & Solomon, 1993; Waha et al., 2012). However, in the humid tropics, where temperature and moisture conditions are suitable for crop growth throughout the year, sowing dates are not well explained by climate alone (Sacks et al., 2010; Waha et al., 2012).

Moreover, even in regions where climate is a determinant of sowing dates, the main climatic constraints vary by region and crop, as exemplified by the relatively greater importance of temperature than precipitation for maize in the American Midwest (Choi et al., 2017), the timing of rainy (dry) season onset for rice-wheat rotation in India (Mathison et al., 2017), the seasonality of monsoons, flooding and salinity intrusion for triple rice cropping in the Vietnamese Mekong Delta (Kotera et al., 2014), and soil moisture levels for millet and sorghum in the dry tropics (Marteau et al., 2011).

Our study confirms the importance of fulfilling the requirements for heat, chilling, and moisture and underpins, for the first time, the importance of considering field workability and multicropping patterns in determining sowing dates (and harvesting dates). Because these factors were rarely considered in earlier work, our findings contribute to the knowledge base for better sowing date modeling.

4.3. Limitations

This study has several limitations. First, the modeled calendars are estimates and should not be considered as "ground truth." The model outputs represent average calendars over a large spatial domain (0.5°). Discrepancies with local calendars could occur. Topography, soil characteristics, the availability of labor,

field size, and access to technology (e.g., use of greenhouse for rice seedling in cooler regions) and information (e.g., weather forecasts) contribute to local calendars (Tian et al., 2014; van der Velde et al., 2012; Yegbemey et al., 2014). Second, day length is important for photoperiod-sensitive varieties in determining their crop duration (Sawano et al., 2008) but was not considered here because global modeling remains a challenge (van Bussel et al., 2015). Third, rice transplanting dates in regions where farmers experience severe typhoon damage are often scheduled to complete harvesting before the main typhoon season. Typhoons, flooding, and salinity intrusion are known seasonal phenomena that damage crop production in Asia (Iizumi et al., 2008; Koide et al., 2013; Kotera et al., 2014; Masutomi et al., 2012) but are not considered here. Fourth, the specification of field workability requirements could be improved. Time periods required to lower grain moisture content before harvesting (Sacks & Kucharik, 2011) and those required to dry farm fields after soil melting (Kucharik, 2006) need to be considered. Planting methods differ by country (hand sowing to manual planting machines to airplane planting), which poses further complications. Fifth, modeled calendars may be sensitive to the choice of meteorological forcing data set, soil data set, and soil water balance model. Lastly, we selected to model crop calendars around the year 2000 to enable us comparing with reported data. Detection and attribution of crop calendar change for the last decades are feasible and interesting, but beyond the scope of the present study.

5. Conclusions

In this paper, we presented a rule-based crop sowing and harvesting windows model. The model reveals that field workability and multicropping patterns are as important as the requirements for heat, chilling, and moisture in determining sowing and harvesting dates. Because previous sowing date and phenology models rarely considered these factors, our findings contribute to better crop modeling, monitoring, and forecasting.

Using the model, we depicted spatially detailed global patterns of sowing and harvesting windows for major crops. Maize, rice, wheat, and soybeans were covered. The distinctions between winter and spring wheat, irrigated and rainfed conditions, and representative varieties were made. Crop calendars are available over the global land area. Although limited crops were considered here, the model considers some major differences between crops (spring vs. wheat and irrigated vs. rainfed), and the model output overcomes some known limitations and shortfalls of earlier crop calendar data sets (i.e., spatial resolution, more complete distinction of cropping season and cropping system than before, and inclusion of cultivar characteristics). The model can be adapted to other crops and time periods with minor modifications.

The daily probabilities of sowing and harvesting derived using the model can be converted into sowing and harvesting windows, respectively, with user-defined thresholds. Multicropping patterns can be resolved with the model outputs. The model outputs are especially valuable in regions where crop calendars are sparse or unreliable and offer opportunities to study multicropping patterns and agricultural intensification.

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